# Practice workbook · Interpret antibacterial culture investigations

Petri dish and growth-graph investigations of antibiotics or bacterial growth: comparing inhibition zones or counts, variables and safe incubation.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ANTIBACTERIAL-CULTURE-INVESTIGATION

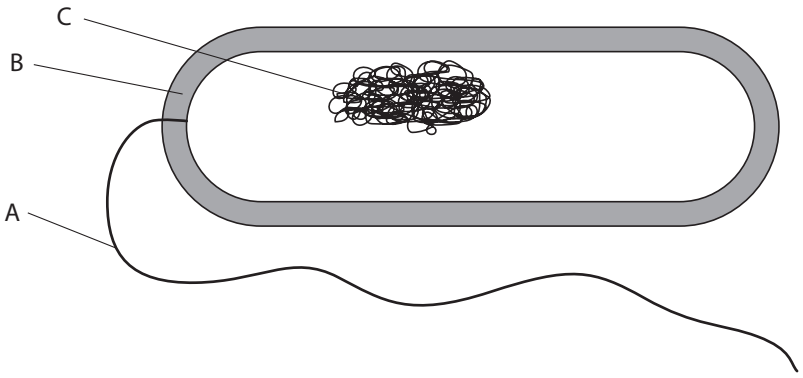
# 1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

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**8** The diagram shows a bacterial cell.




(a) (i) Name the parts labelled A, B and C.

**(3)**

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

**(1)**

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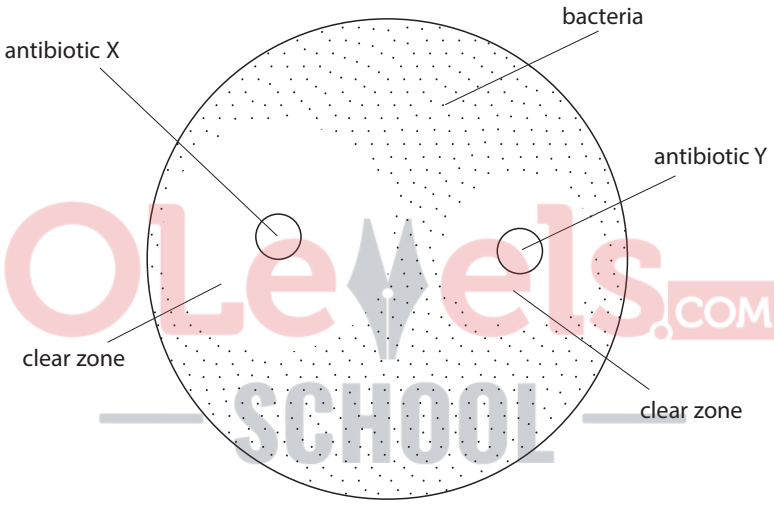
# 1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days




(i) Describe what is meant by the term **antibiotic**.

**(2)**

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# 1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

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(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

**(3)**

(iii) Explain a suitable control that could be used in this investigation.

**(3)**

**(Total for Question 8 = 12 marks)**

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## 1. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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## 2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

**7** Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

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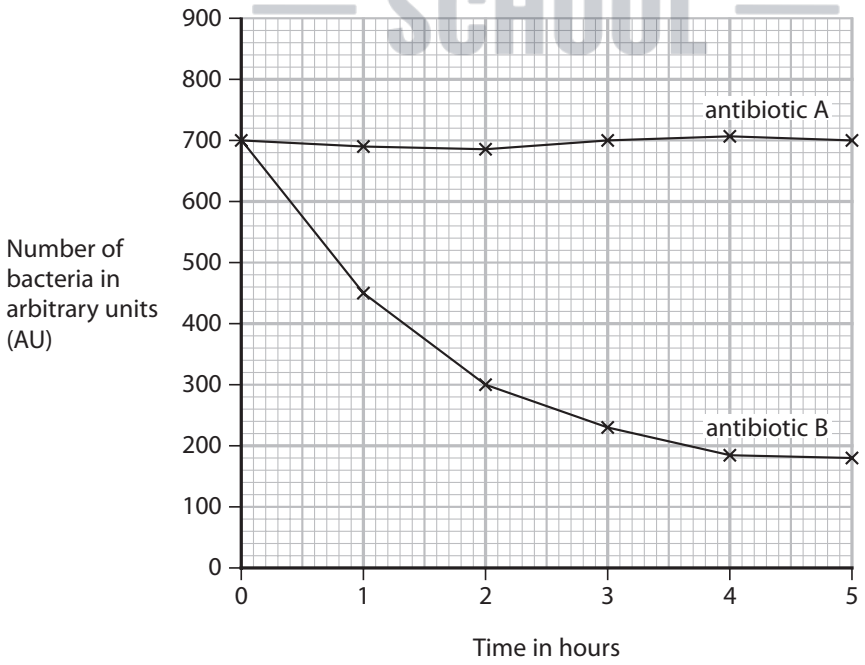
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.




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## 2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

**(3)**

mean rate of change = ..... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

**(2)**

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## 2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(iii) Discuss which antibiotic would be better to use to treat any disease caused by the bacteria.

**(3)**

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**(Total for Question 7 = 11 marks)**

Working and answer

## 3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

**7** Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

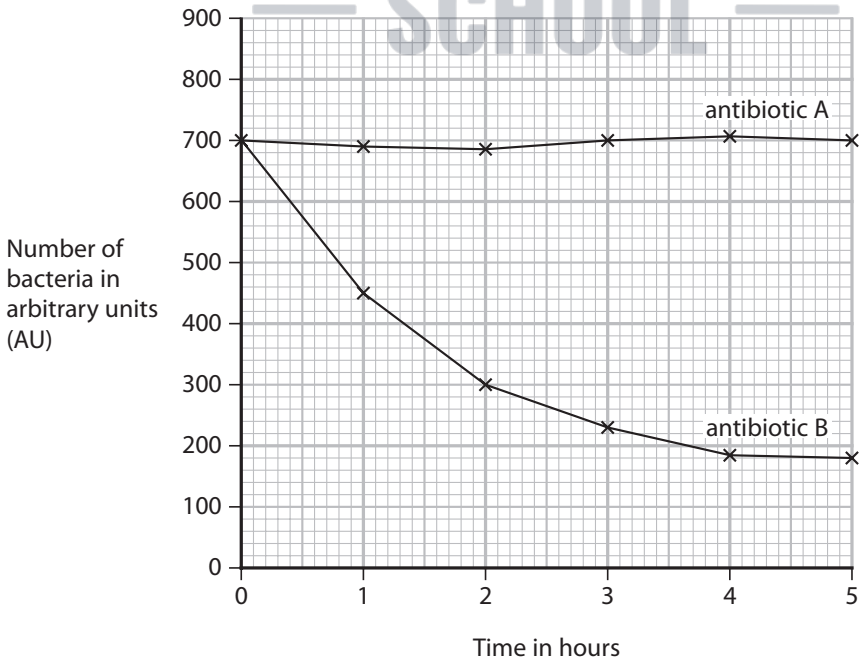
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.




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## 3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

**(3)**

mean rate of change = ..... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

**(2)**

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## 3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

(iii) Discuss which antibiotic would be better to use to treat any disease caused by the bacteria.

**(3)**

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**(Total for Question 7 = 11 marks)**

Working and answer

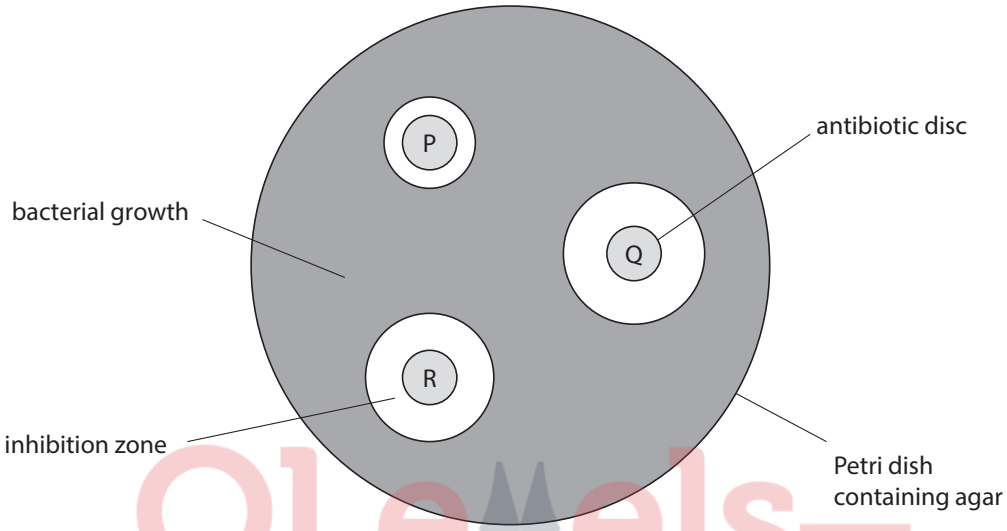
# 4. Question 5 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

**5** (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.




(i) Which is the independent variable in this investigation?

**(1)**

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

**(2)**

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# 4. Question 5 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

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(iii) Explain which is the most effective antibiotic.

**(2)**

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

**(4)**

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(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

**(2)**

**(Total for Question 5 = 11 marks)**

## 4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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# 5. Question 7 · part b.ii · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

**7** Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 $cm^3$ of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 $cm^3$ of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

| Petri dish | Diameter of bacterial growth in mm | | | | |
|---|---|---|---|---|---|
| | day 1 | day 2 | day 3 | day 4 | day 5 |
| A | 0 | 1.7 | 3.0 | 4.6 | 7.1 |
| B | 0 | 4.2 | 8.4 | 15.1 | 36.5 |
| C | 0 | 0 | 0 | 0 | 0 |

(a) (i) State three control variables in the student's investigation.

**(3)**

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2 .....

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3 .....

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## 5. Question 7 · part b.ii · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

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(ii) State the dependent variable in the student's investigation.

**(1)**

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

**(2)**

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

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(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

**(4)**

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

**(2)**

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## 5. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

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